Prior art: who has built which piece

Component matrix for *critical Dicke electrometry*

Extending intracavity Rydberg-EIT microwave sensing (Peng *et al.*, JOSA B **35**, 2272) into the superradiant phase transition (Zhang *et al.*, PRL **110**, 090402). August 24, 2026

Legend: ✓ present – absent –/✓ *the cell that decides the strand.* **CRT** = counter-rotating light–matter terms (their presence is what permits a $\mathbb{Z}_2$ superradiant transition). **MW = measurand** means the microwave field is the quantity being measured, not merely a drive.

Paper	EIT	Cav.	$g\sqrt{N}$	CRT	Transition	Ryd. int.	MW = meas.	Th/Exp	Headline result
A. Cavity + Rydberg electrometry everything but the counter-rotating terms ⇒ no transition at all									
Peng 2018 JOSA B 35, 2272 ★	✓	✓	– weak	–	–	–	✓	Th	8× lower detectable field; >20× resolution
Yang 2020 JOSA B 37, 1664	✓	✓	✓ coll. Rabi	–	–	–	✓	Th	7× over plain EIT; four-peak transmission
Wang 2023 JOSA B 40, 2604	✓	✓	✓ strong	–	–	–	✓	Th	196.7× / 26.2×; floor 396.5 nV cm^{-1}
Jing 2020 Nat. Phys. 16, 911	✓	–	–	–	–	–	✓ superhet	Exp	55 nV cm^{-1}Hz$^{-1/2}$
Liu 2025 Chin. Phys. Lett. 42, 053201	✓	– *MW* cav.	–	–	–	–	✓	Exp	18 dB power sensitivity
arXiv:2502.20792 cavity superhet	✓	✓	✓	–	–	–	✓ superhet	Exp	19 dB sensitivity gain
B. Critical Rydberg electrometry has criticality *and* sensing — but the interaction-driven transition, not light–matter									
Ding 2022 Nat. Phys. 18, 1447	✓	–	–	–	✓ NEPT / bistability	✓ drives it	✓	Exp	FI ×10^3 vs. independent particles; 49 nV cm^{-1}Hz$^{-1/2}$
Wang 2025 arXiv:2502.19761	✓	✓	–	–	✓ NEPT / bistability	✓ drives it	✓	Exp	classical FI ×100 from cavity; 2.6 nV cm^{-1}Hz$^{-1/2}$
C. Dicke model / phase structure has the transition, the cavity and the interactions — and senses nothing									
Dimer 2007 / Baden 2014 PRA 75, 013804 / PRL 113, 020408	–	✓	✓	✓ Raman	✓ $\mathbb{Z}_2$ Dicke	–	–	Th / Exp	Engineered Dicke QPT via cavity-assisted Raman
Zhang 2013 PRL 110, 090402 ★	2-photon	✓	✓	– **RWA**	✓ $U(1)$ polariton + SRS	✓	–	Th (QMC)	Superradiant solid; first-order boundaries
arXiv:2503.13949 anisotropic engineering	–	✓	✓	✓ tunable $0\to\infty$	✓	✓	–	Th	Periodic-driving recipe for the anisotropic Dicke model
arXiv:2511.22230 Dicke–Ising QMC	–	✓	✓	✓	✓ SR / SRS / Solid-$\frac{1}{2}$	✓	– MW = *drive*	Th (QMC)	Orders of all transitions; blockade suppresses cavity occupation
Garbe 20/22, Rams 18, Gietka 21/22, Gyhm 25	–	✓	✓	✓	✓	–	– generic param.	Th	Critical-metrology scaling — and its no-gos
arXiv:1611.00797 steady-state SR with Rydberg polaritons	–	✓	✓	–	✓ lasing threshold	✓	–	Th	Rydberg nonlinearity strongly modifies the superradiance threshold
D. This proposal the same protocol as strand B, with a *light–matter* transition in place of the interaction-driven one									
Route I — $U(1)$ lasing after Zhang 2013	✓	✓	✓	– + repump	✓ $U(1)$ lasing	optional	✓ **sets** G^2_{th}	Th	cavity drift rejected 101×; 0.070 nV cm^{-1}Hz$^{-1/2}$ at N=10^6
Route II — $\mathbb{Z}_2$ Dicke after Dimer/Baden	✓	✓	✓	✓	✓ $\mathbb{Z}_2$ Dicke	optional	✓ **sets** λ_c	Th	ε^{-1} gain; −10.3 dB self-squeezing

★ = the two seed papers this project starts from.

Full discussion in `dicke_rydberg_sensing.pdf`, §6. Reading notes and references overleaf.

Reading the matrix

Every individual column is occupied, and each strand is exactly *one column* short of the proposal.

- **Strand A** lacks counter-rotating terms, so there is no transition at all. It stops at a collective Rabi splitting read out spectroscopically.
- **Strand B** has criticality and sensing, but its critical point is set by atomic density and interaction strength: it drifts with temperature and stray fields, it is hysteretic (the discriminator depends on scan direction), and it cannot be tuned quickly.
- **Strand C** has the light–matter transitions and does no sensing. Two near-misses: arXiv:2511.22230, where microwave fields are *in* the setup but as the drive that engineers the model rather than the measured quantity; and arXiv:1611.00797, which puts a Rydberg medium at a superradiance threshold in a cavity, but to generate nonclassical light.
- **Strands B and D are the same phase diagram.** Turning up the Rydberg interaction in Route I folds its threshold over into optical bistability — i.e. back into strand B. The proposal is a move between corners of one model, not a competing claim.

The empty cell that defines the project is the intersection of the last two columns — a *light–matter* transition whose phase boundary *is* the measurand. Both flavours depend on the same dressed splitting, so both are sensing curves:

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2\,, \qquad \Omega_\mu = dE_\mu/\hbar\,; \qquad G_{\rm th}^2 = \kappa\Gamma\left[1 + \frac{(\omega_{\rm cav}-\tilde{\omega}_0)^2}{(\kappa+\Gamma)^2}\right] \;\; (U(1)), \qquad \lambda_c = \tfrac{1}{2}\sqrt{\omega\,\tilde{\omega}_0} \;\; (\mathbb{Z}_2).$$

Two cells worth staring at

- **Zhang 2013 → – (RWA).** Their coupling $(g/\sqrt{N})\sum_i(b_i^\dagger a_i\psi + \text{H.c.})$ conserves excitation number — which is why their phase diagram is drawn against a chemical potential. Their "generalised Dicke model" is therefore of Tavis–Cummings type and their superradiance is a $U(1)$ polariton condensation, not the $\mathbb{Z}_2$ transition this proposal needs. *The seed paper is one column short of the required model, so the extension is real rather than a re-derivation.*
- **arXiv:2503.13949 → ✓ (tunable $0\to\infty$).** The enabling technology, already published — and it fills exactly the cell Zhang 2013 leaves empty, for the Rydberg platform specifically.

The same RWA restriction runs through the whole of strand A. **Counter-rotating terms are the single ingredient separating this proposal from that entire lineage.**

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